

Linear Models Lecture 10: Asymptotics

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Big Picture

- We began with a model in which $Y = \mathbf{x}'\boldsymbol{\beta} + e$
- OLS finds the projection of $\mathbf{y}$ in the vector space spanned by $\mathbf{x}$:

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = \mathbf{P}\mathbf{y}$$

$$\hat{\mathbf{e}} = \mathbf{y} - \hat{\mathbf{y}} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{y} - \mathbf{P}\mathbf{y} = (\mathbf{I} - \mathbf{P})\mathbf{y} = \mathbf{M}\mathbf{y}$$

- We showed that the projection coefficient $\hat{\boldsymbol{\beta}}$ is an unbiased estimator of $\boldsymbol{\beta}$.
- With the additional assumption of normality, we derived t and F distributions (Lecture 7).

In Lecture 9, we used asymptotic arguments **informally**: Taylor expansion of the score, LLN for the Hessian, CLT for the score. Today we develop these tools rigorously.

Standards for Inference

- Unbiased estimators ($E[\hat{\theta}] = \theta$) are “correct on average” even in small samples.
 - Unbiasedness tells us that we are not better off with a different guess (we would lose money in expectation).
 - A biased estimator is biased even with infinite data.
- Unbiasedness is not always what we want.
 - Unbiasedness doesn't rule out pathological behavior: the “first impression” estimator is unbiased.
 - Many popular estimators are biased: Probit MLE (Lecture 9), 2SLS, ridge regression.
- A biased estimator may be a good approximation if it gets **closer to the truth with more data**.
- What is a “good” approximation?

Why Consistency Matters: A Concrete Example

U.S. hourly wages: $\mu = 24$, $\sigma^2 = 430$.

“First Impression” estimator

$\hat{\mu} = X_1$ (use only the first observation)

- Unbiased: $\mathbb{E}[X_1] = \mu = 24$
- $\text{Var}(\hat{\mu}) = 430$ for *any* n
- 95% interval: 24 ± 40.7

Sample mean $\bar{X}_n$

- Unbiased: $\mathbb{E}[\bar{X}_n] = \mu = 24$
- $\text{Var}(\bar{X}_n) = 430/n \rightarrow 0$
- 95% interval at $n = 100$: 24 ± 4.1

Both estimators are unbiased. Only $\bar{X}_n$ is **consistent**: its distribution collapses onto μ .
When forced to choose, **consistency** > **unbiasedness**.

It has been customary to assume (somewhat loosely) that when a quantity is calculated from a random sample to estimate a parameter of a hypothetical frequency distribution, the accuracy of the determination will increase without limit as the number in the sample increases. A *consistent statistic* is a function of the observations actually possessing this property.

- Hotelling 1930

Consistency

- Consistent estimators are “more likely correct with more data”.
- The analog to expectations in large samples is called the probability limit: plim .
- Claim of Consistency: $\text{plim} \hat{\beta} = \beta$, or $\hat{\beta}_j$ is *consistent* for β_j

Convergence of Random Variables

Consider the sample mean $\bar{X}_n$ as a sequence indexed by sample size n . Its distribution changes with n —how do we describe its limit?

There is a hierarchy of convergence concepts for random variables:

$$X_n \xrightarrow{\text{a.s.}} X \quad \Rightarrow \quad X_n \xrightarrow{p} X \quad \Rightarrow \quad X_n \xrightarrow{d} X$$

- **Almost sure (a.s.) convergence** is the strongest form—pathwise convergence except on a probability-zero set. We won't need it operationally.
- **Convergence in probability** ($\xrightarrow{p}$): for all $\varepsilon > 0$, $\mathbb{P}(|X_n - X| > \varepsilon) \rightarrow 0$.
 - This is the concept behind **consistency** (WLLN, plim, CMT).
- **Convergence in distribution** ($\xrightarrow{d}$): $F_{X_n}(t) \rightarrow F_X(t)$ at all continuity points.
 - This is the concept behind the **CLT** and asymptotic normality.

Today we develop each in turn: $\xrightarrow{p}$ (next), then $\xrightarrow{d}$ (CLT section).

Convergence in Probability

A sequence of random variables X_n converges *in probability* to c if, for all $\delta > 0$,

$$\lim_{n \rightarrow \infty} Pr(|X_n - c| \leq \delta) = 1$$

This is written as $X_n \xrightarrow[p]{p} c$ or $\text{plim} X_n = c$

We can also say

$$\lim_{n \rightarrow \infty} Pr(|X_n - c| > \delta) = 0$$

That is, the random variable concentrates about a point in a certain interval.

Plim example

- Define a random variable Z_n that takes on values 0 and a_n with probability $1 - p_n$ and p_n respectively,
- Intuitively Z_n should converge to 0 if either $p_n \rightarrow 0$ or $a_n \rightarrow 0$.
- Consider any $\delta > 0$. If $a_n \rightarrow 0$, then for n large enough, $a_n < \delta$. If so $Pr(|Z_n| \leq \delta) = 1$.
- If $p_n \rightarrow 0$ then $1 - p_n \rightarrow 1$.
- $1 - p_n$ of the time $Z_n = 0 < \delta$, so $Pr(|Z_n| \leq \delta) \geq 1 - p_n$
- Therefore $Pr(|Z_n| \leq \delta) \rightarrow 1$ as $p_n \rightarrow 0$

Consistency in Practice: OLS with Increasing n

DGP: $Y_i = 2 + 3X_i + \varepsilon_i$, $X_i \sim N(0,1)$, $\varepsilon_i \sim \text{Exp}(1) - 1$ (skewed, non-normal errors).

```
set.seed(42)
par(mfrow = c(2, 2))
for (n in c(20, 50, 200, 2000)) {
  betas <- replicate(5000, {
    x <- rnorm(n); y <- 2 + 3*x + (rexp(n)-1)
    coef(lm(y ~ x))[2]
  })
  hist(betas, breaks = 40, prob = TRUE,
       main = paste("n =", n), xlim = c(2, 4),
       xlab = expression(hat(beta)[1]))
  abline(v = 3, col = "blue", lwd = 2)
}
```

Errors are **skewed and non-normal**, yet $\hat{\beta}_1$ concentrates around the true value $\beta_1 = 3$ as n grows. This is consistency in action.

Why Useful : The continuous mapping theorem

- Unlike expectations, plim survives continuous functions (like division!).

The continuity theorem, the first of Slutsky's theorems, the continuous mapping theorem.

Given a random variable x_n that converges in probability to x and a continuous function $g(x_n)$, we have that:

$$\text{plim} g(x_n) = g(\text{plim} x_n)$$

If $\text{plim} x_n = c$ and $\text{plim} y_n = d$, then

$$\text{plim}(x_n y_n) = (\text{plim} x_n)(\text{plim} y_n) = cd$$

$$\text{plim}(x_n / y_n) = (\text{plim} x_n) / (\text{plim} y_n) = c/d$$

$$\text{plim}(x_n + y_n) = (\text{plim} x_n) + (\text{plim} y_n) = c + d$$

CMT in Action: Why Division Works

The t -statistic divides one random quantity by another. Why is this valid?

- WLLN: $\bar{X}_n \xrightarrow{P} \mu$ and $s_n^2 \xrightarrow{P} \sigma^2$
- CMT (continuous function $g(a, b) = a/\sqrt{b}$):

$$\frac{\bar{X}_n}{s_n} \xrightarrow{P} \frac{\mu}{\sigma}$$

- More generally, any smooth function of consistent estimators is itself consistent.

Every time you divide an estimate by its standard error, you rely on CMT. Without it, ratios of random variables have no guarantee of converging to the right thing.

Weak Law of Large Numbers

Proof plan:

- Markov's Inequality
- Chebyshev's Inequality
- Weak Law of Large Numbers

Markov's Inequality

Proposition

Suppose X is a random variable that takes on non-negative values. Then, for all $a > 0$,

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Markov's Inequality

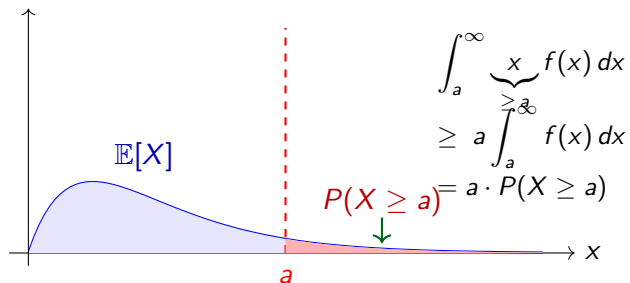
Proof.

For $a > 0$,

$$\begin{aligned} E[X] &= \int_0^{\infty} xf(x)dx \\ &= \int_0^a xf(x)dx + \int_a^{\infty} xf(x)dx \\ E[X] &\geq \int_a^{\infty} xf(x)dx \geq \int_a^{\infty} af(x)dx = aP(X \geq a) \quad (X \geq 0) \\ \frac{E[X]}{a} &\geq P(X \geq a) \end{aligned}$$



Markov's Inequality: Visual Intuition



The full integral $\mathbb{E}[X]$ includes the **tail area**. In the tail, every $x \geq a$, so replacing x with a only makes the integral smaller. Hence $\mathbb{E}[X] \geq a \cdot P(X \geq a)$.

Chebyshev's Inequality

Proposition

If X is a random variable with mean μ and variance σ^2 , then, for any value $k > 0$,

$$P(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2}$$

Chebyshev's Inequality

Proof.

Define $Y = (X - \mu)^2$ where $\mu = E[X]$. Then $Y \geq 0$.

Set $a = k^2$ in Markov's inequality:

$$\begin{aligned} P(Y \geq k^2) &\leq \frac{E[Y]}{k^2} \\ P((X - \mu)^2 \geq k^2) &\leq \frac{E[(X - \mu)^2]}{k^2} = \frac{\sigma^2}{k^2} \\ P(|X - \mu| \geq k) &\leq \frac{\sigma^2}{k^2} \end{aligned}$$



Example of Markov and Chebyshev

Suppose that we know that the number of items produced in a factory is a random variable with mean 500.

- How likely are we to see a production of 1000?
- If the variance is 50, how likely is the production to be between 400 and 600?

Solution

- How likely are we to see a production of 1000?

Markov's Inequality:

$$\mathbb{P}(X \geq 1000) \leq \frac{\mathbb{E}[X]}{1000} = \frac{500}{1000} = 0.5$$

- How likely is production between 400 and 600, given variance = 50?

Chebyshev's Inequality:

$$\mathbb{P}(|X - 500| \geq 100) \leq \frac{50}{100^2} = \frac{1}{200}$$

$$\mathbb{P}(400 \leq X \leq 600) = \mathbb{P}(|X - 500| < 100) \geq 1 - \frac{1}{200} = \frac{199}{200}$$

Chebyshev Sample Size Calculation

- U.S. hourly wages are distributed $\mu = 24$ and $\sigma^2 = 430$,
- How large does n have to be so that $\bar{X}_n$ is within \$1 of the true mean with probability 99%?

$$\mathbb{P} [|\bar{X}_n - \mu| \geq 1] = \mathbb{P} [|\bar{X}_n - \mathbb{E} [\bar{X}_n]| \geq 1] \leq \text{var} [\bar{X}_n] = 430/n$$
$$\frac{430}{n} \leq .01$$

$$\mathbb{P} [|\bar{X}_n - \mu| \geq 1] \leq .01 \text{ if } n = 43,000$$

Weak Law of Large Numbers

Proposition

Suppose $X_1, X_2, \dots, X_n$ is iid random sample from a distribution with expectation μ and $\text{Var}(X_i) = \sigma^2$. Then, for all $\varepsilon > 0$,

$$P \left\{ \left| \frac{X_1 + X_2 + \dots + X_n}{n} - \mu \right| \geq \varepsilon \right\} \rightarrow 0 \text{ as } n \rightarrow \infty$$

or

$$\bar{X}_n \xrightarrow{p} \mathbb{E}[X]$$

Weak Law of Large Numbers

Proof.

$$\frac{\mathbb{E}[X_1 + X_2 + \cdots + X_n]}{n} = \frac{\sum_{i=1}^n \mathbb{E}[X_i]}{n} = \mu$$

$$\mathbb{E} \left[\left(\frac{\sum_{i=1}^n X_i - n\mu}{n} \right)^2 \right] = \frac{\text{Var}(X_1 + X_2 + \cdots + X_n)}{n^2} = \frac{\sum_{i=1}^n \text{Var}(X_i)}{n^2} = \frac{\sigma^2}{n}$$

$$\mathbb{P} \left(\left| \frac{X_1 + X_2 + \cdots + X_n}{n} - \mu \right| \geq \varepsilon \right) \leq \frac{\sigma^2}{n\varepsilon^2} \rightarrow 0 \quad \text{as } n \rightarrow \infty$$



WLLN interpretation

- The sample mean approaches the population mean as $n \rightarrow \infty$
- General version only assumes $\mathbb{E}[X] < \infty$
- This notion that a sequence of the same sample quantity approaches a constant is called consistency.

Any sample moment converges to its population counterpart: $\frac{1}{n} \sum_{i=1}^n X_i X_i' \xrightarrow{P} \mathbb{E}[X_i X_i']$, $\frac{1}{n} \sum_{i=1}^n X_i Y_i \xrightarrow{P} \mathbb{E}[X_i Y_i]$. This is the foundation for OLS consistency (Lecture 11).

WLLN Counterexample

- WLLN needs to have independent observations:
- For example, assume $X_i = Z + U_i$, like in the random effects model.
- Suppose Z and U_i are random and independent, $\mathbb{E}(U_i) = 0$.
- $\bar{X}_n = Z + \bar{U}_n \xrightarrow[p]{} Z$ not a constant.

Big O and Little o : The Idea

When we take limits, we often need to say “this term is negligible compared to that one.” Asymptotic notation gives us a precise language for this.

- Consider $f(n) = 3n^2 + 5n + 7$ as $n \rightarrow \infty$.
- For large n , the $3n^2$ term dominates: at $n = 100$, we have $30,000 + 500 + 7$.
- We want to say: “ $f(n)$ grows like n^2 ” and “the $5n + 7$ part is *negligible* relative to n^2 .”

Two notions of “negligible”:

- **Little o** = “*vanishes* relative to”: a_n shrinks *faster* than b_n
- **Big O** = “*bounded* relative to”: a_n grows *no faster* than b_n

Analogy: if b_n is a speed limit, then $a_n = o(b_n)$ means a_n is slowing to a stop, while $a_n = O(b_n)$ means a_n is obeying the speed limit (but may still be driving).

Formal Definitions: O and o

Definition

Let $\{a_n\}$ and $\{b_n\}$ be sequences with $b_n > 0$.

■ **Little o :** $a_n = o(b_n)$ means $\frac{a_n}{b_n} \rightarrow 0$ as $n \rightarrow \infty$.

■ **Big O :** $a_n = O(b_n)$ means $\left| \frac{a_n}{b_n} \right| \leq C$ for some constant C and all n large enough.

In words:

■ $a_n = o(b_n)$: “ a_n vanishes compared to b_n ” (the ratio goes to zero)

■ $a_n = O(b_n)$: “ a_n is at most on the order of b_n ” (the ratio stays bounded)

Note: little o is a *stronger* statement than big O . If $a_n = o(b_n)$ then $a_n = O(b_n)$, but not vice versa.

Examples: O and o with Sequences

Statement	Check	Why
$5n = o(n^2)$	$5n/n^2 = 5/n \rightarrow 0 \checkmark$	n grows slower than n^2
$5n = O(n^2)$	$5n/n^2 = 5/n \leq 5 \checkmark$	also bounded (weaker)
$3n^2 = O(n^2)$	$3n^2/n^2 = 3 \checkmark$	bounded but doesn't vanish
$3n^2 = o(n^2)?$	$3n^2/n^2 = 3 \not\rightarrow 0 \times$	same order, not negligible
$1/n = o(1)$	$\frac{1/n}{1} \rightarrow 0 \checkmark$	$1/n$ vanishes
$1/n^2 = o(1/n)$	$\frac{1/n^2}{1/n} = 1/n \rightarrow 0 \checkmark$	$1/n^2$ vanishes faster

The punchline: When we write $f(x) = g(x) + O(x^3)$, the leftover terms are *at most* on the order of x^3 . When x is small (like $t/\sqrt{n}$), these terms are negligible.

Application: Taylor Remainder in O Notation

Consider the Taylor expansion of $\ln(1+x)$ around $x=0$:

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$$

We can truncate and describe what we dropped:

$$\ln(1+x) = x + O(x^2) \quad (\text{first-order: remainder} \leq Cx^2)$$

$$\ln(1+x) = x - \frac{x^2}{2} + O(x^3) \quad (\text{second-order: remainder} \leq Cx^3)$$

Why this is useful: If $x = t/\sqrt{n}$, then:

- $O(x^2) = O(1/n)$ — vanishes as n grows
- $O(x^3) = O(n^{-3/2})$ — vanishes even faster

This is exactly how we will control the error in the CLT proof: the Taylor remainder in the MGF expansion becomes negligible as $n \rightarrow \infty$.

Big O_p and Little o_p : The Stochastic Versions

For *random* variables, we add a subscript p (“in probability”):

Definition

- $X_n = o_p(a_n)$: $X_n/a_n \xrightarrow{p} 0$ (“*vanishes in probability*”)
- $X_n = O_p(a_n)$: X_n/a_n *stays bounded in probability*

Translations to what we just proved:

- $X_n \xrightarrow{p} c$ is the same as writing $X_n - c = o_p(1)$
- WLLN says $\bar{X}_n - \mu = o_p(1)$, and more precisely $\bar{X}_n - \mu = O_p(n^{-1/2})$
- CLT (coming next) says $\sqrt{n}(\bar{X}_n - \mu) = O_p(1)$

When you see $o_p(1)$ in a proof, read it as “a term that vanishes in probability — we can ignore it.”

What is a Taylor Expansion?

Problem: We know everything about a function f at a point a (its value, slope, curvature, ...), and we want to approximate f at a *nearby* point b .

Idea: Build the approximation one piece at a time:

- 1 Zeroth order:** Start with $f(b) \approx f(a)$ (constant — ignore everything)
- 2 First order:** Add the slope: $f(b) \approx f(a) + f'(a)(b - a)$ (tangent line)
- 3 Second order:** Add the curvature: $f(b) \approx f(a) + f'(a)(b - a) + \frac{f''(a)}{2}(b - a)^2$
- 4** Keep going: each term uses one more derivative and improves the fit near a .

Key insight: The closer b is to a , the better the approximation. If $b - a$ is small, then $(b - a)^2$ is *very* small, $(b - a)^3$ is *tiny*, etc. — so the first few terms do most of the work.

Taylor Approximation: First Order

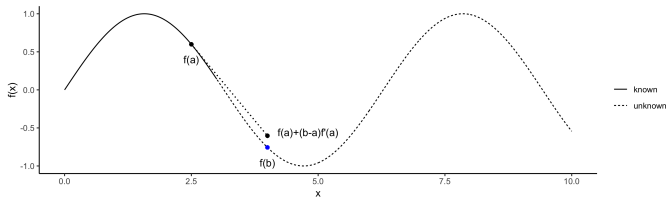
- We know $f(a)$ and $f'(a)$. We want to approximate $f(b)$ for b near a .
- **First-order Taylor expansion:**

$$f(b) = f(a) + f'(a)(b - a) + R_1(b)$$

where the remainder $R_1(b) = \frac{f''(c)}{2}(b - a)^2$ for some c between a and b .

- In asymptotic-order notation:

$$f(b) = f(a) + f'(a)(b - a) + O((b - a)^2)$$



Taylor Approximation: Higher Orders

The k -th order Taylor expansion of f around a :

$$f(b) = \sum_{j=0}^k \frac{f^{(j)}(a)}{j!} (b-a)^j + O((b-a)^{k+1})$$

Second-order expansion (used constantly in asymptotic theory):

$$f(b) = f(a) + f'(a)(b-a) + \frac{f''(a)}{2}(b-a)^2 + O((b-a)^3)$$

Why this matters:

- The **Delta Method** uses the first-order expansion to linearize functions of estimators.
- The **CLT proof** uses the second-order expansion of the moment generating function.
- The $O(\cdot)$ notation tells us exactly which terms we can drop as $n \rightarrow \infty$.

The Exponential Function as a Series

The most important function in mathematics is $\exp(x)$, *defined* by:

$$\exp(x) = \sum_{k=0}^{\infty} \frac{x^k}{k!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

Key properties (all follow from the series):

- 1 $\exp(a)\exp(b) = \exp(a+b)$, so we write $e^x \equiv \exp(x)$ where $e = \exp(1) = 2.718\dots$
- 2 $\frac{d}{dx}e^x = e^x$: differentiate term-by-term and re-index.
- 3 The series *is* the Taylor expansion of e^x around 0 (since every derivative is e^x and $e^0 = 1$).

Why this matters now: We will expand $\mathbb{E}[e^{tX}]$ using this series to extract moments, and then use the Taylor remainder to control the error in the CLT proof.

The Compound Interest Limit

Proposition

$$\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x$$

Proof: Take logs, apply the Taylor expansion $\ln(1 + u) = u - \frac{u^2}{2} + O(u^3)$:

$$n \ln\left(1 + \frac{x}{n}\right) = n\left(\frac{x}{n} - \frac{x^2}{2n^2} + O(n^{-3})\right) = x - \frac{x^2}{2n} + O(n^{-2}) \rightarrow x$$

So $\left(1 + \frac{x}{n}\right)^n = e^{n \ln(1+x/n)} \rightarrow e^x$.

This converts a product of n nearly-identical terms into an exponential — the crucial step in the CLT proof: $(1 + t^2/(2n) + \text{small})^n \rightarrow e^{t^2/2}$.

Moment Generating Functions

Definition

The **moment generating function** (MGF) of a random variable X is:

$$M_X(t) = \mathbb{E}[e^{tX}]$$

defined for all t in a neighborhood of 0.

Why “moment generating”? Expand e^{tX} using the series definition:

$$\begin{aligned} M_X(t) &= \mathbb{E}\left[\sum_{k=0}^{\infty} \frac{(tX)^k}{k!}\right] = \sum_{k=0}^{\infty} \frac{t^k}{k!} \mathbb{E}[X^k] \\ &= 1 + t \mathbb{E}[X] + \frac{t^2}{2} \mathbb{E}[X^2] + \dots \end{aligned}$$

So $M_X^{(k)}(0) = \mathbb{E}[X^k]$: the k -th derivative at zero gives the k -th moment.

MGFs: Key Properties

Why MGFs are useful for proving limit theorems:

- 1 Uniqueness:** If $M_X(t) = M_Y(t)$ in a neighborhood of 0, then X and Y have the same distribution.
- 2 Independence:** If X and Y are independent, $M_{X+Y}(t) = M_X(t) \cdot M_Y(t)$.
 - For n iid copies: $M_{X_1+\dots+X_n}(t) = [M_{X_1}(t)]^n$
- 3 Continuity theorem:** If $M_{X_n}(t) \rightarrow M_X(t)$ for all t near 0, then $X_n \xrightarrow{d} X$.

Standard normal MGF: If $Z \sim N(0, 1)$, then $M_Z(t) = e^{t^2/2}$.

Proof strategy for CLT: Show that the MGF of the standardized sample mean converges to $e^{t^2/2}$.

Convergence in Distribution

- We want the sampling distribution of the sample mean $\bar{X}_n$.
- The sampling distribution is a function of the distribution of observations F and the sample size n .
- We will get an *asymptotic approximation* by standardizing $\bar{X}_n$ and taking the limit as $n \rightarrow \infty$

Definition: Convergence in Distribution

- Let X_n be a sequence of random variables, and X a target random variable.
- We say that X_n **converges in distribution** to X if:

$$\lim_{n \rightarrow \infty} \mathbb{P}(X_n \leq t) = \mathbb{P}(X \leq t) \quad \text{for all } t \text{ where } F_X(t) \text{ is continuous}$$

- **Implications**

- The cumulative distribution functions converge: $F_{X_n}(t) \rightarrow F_X(t)$
- If the target distribution is a constant, then this is just convergence in probability

- **Notation:** $X_n \xrightarrow{d} X$

Standardizing the Sample Mean

- The WLLN tells us that $\bar{X}_n \xrightarrow{p} \mu$, so $\bar{X}_n \xrightarrow{d} \mu$.
- But the distribution of μ is degenerate, not useful.
- So rescale by the variance: $\text{var}[\bar{X}_n - \mu] = \frac{\sigma^2}{n}$, so $\text{var}[\sqrt{n}(\bar{X}_n - \mu)] = \sigma^2$

$$Z_n = \sqrt{n}(\bar{X}_n - \mu)$$

- As $n \rightarrow \infty$, $\mathbb{E}[Z_n] \rightarrow 0$, $\mathbb{E}[Z_n^2] \rightarrow \sigma^2$, $\mathbb{E}[Z_n^3] \rightarrow 0$, $\mathbb{E}[Z_n^4] \rightarrow 3\sigma^4$
- The Central Limit Theorem states that $Z_n \xrightarrow{d} Z$ where $Z \sim N(0, \sigma^2)$.
- This means that $\bar{X}_n \underset{a}{\sim} N(\mu, \frac{\sigma^2}{n})$.

The Central Limit Theorem

Theorem (Lindeberg–Lévy CLT)

Let $X_1, X_2, \dots$ be iid with $\mathbb{E}[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2 < \infty$. Then:

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2)$$

Equivalently: $\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1)$.

Key conditions:

- **iid** — can be relaxed (Lindeberg–Feller allows heterogeneity)
- **Finite variance** — essential; without it, convergence may fail or be to a non-normal limit

Remarkable fact: The limit distribution is $N(0, \sigma^2)$ **regardless** of the distribution of X_i .

Whether the data are Bernoulli, exponential, or uniform, the sample mean is approximately normal for large n .

Proof of the CLT: Setup

Goal: Show $Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1)$.

Strategy: Show that $M_{Z_n}(t) \rightarrow e^{t^2/2}$, the MGF of $N(0, 1)$.

Step 1. Let $W_i = \frac{X_i - \mu}{\sigma}$, so $\mathbb{E}[W_i] = 0$ and $\mathbb{E}[W_i^2] = 1$. Then:

$$Z_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n W_i$$

Step 2. By independence:

$$M_{Z_n}(t) = \left[M_W\left(\frac{t}{\sqrt{n}}\right) \right]^n$$

We need to evaluate $M_W(t/\sqrt{n})$ for large n , i.e., for small argument. This is where Taylor expansion enters.

Proof of the CLT: Taylor Expansion of the MGF

Step 3. Taylor-expand $M_W(s)$ around $s = 0$, using $M_W^{(k)}(0) = \mathbb{E}[W^k]$:

$$M_W(s) = \underbrace{1}_{M_W(0)} + \underbrace{0}_{E[W]} \cdot s + \frac{1}{2} s^2 + O(s^3) = 1 + \frac{s^2}{2} + O(s^3)$$

Step 4. Substitute $s = t/\sqrt{n}$ and raise to the n -th power (by independence):

$$M_W\left(\frac{t}{\sqrt{n}}\right) = 1 + \frac{t^2}{2n} + O(n^{-3/2})$$
$$M_{Z_n}(t) = \left[M_W\left(\frac{t}{\sqrt{n}}\right)\right]^n = \left[1 + \frac{t^2}{2n} + O(n^{-3/2})\right]^n$$

Proof of the CLT: The Exponential Limit

Step 5. Apply the limit $(1 + x/n)^n \rightarrow e^x$:

$$M_{Z_n}(t) = \left[1 + \frac{t^2/2}{n} + O(n^{-3/2}) \right]^n \rightarrow e^{t^2/2}$$

The $O(n^{-3/2})$ remainder vanishes faster than $1/n$, so it does not affect the limit. The limit $e^{t^2/2}$ is the MGF of $N(0, 1)$.

Step 6. By the continuity theorem for MGFs:

$$Z_n = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} N(0, 1) \quad \blacksquare$$

Taylor expansion $\rightarrow 1 + t^2/(2n) + \text{small}$; exponential limit $\rightarrow e^{t^2/2}$; MGF continuity theorem $\rightarrow$ convergence in distribution.

CLT Sample Size Calculation

- U.S. wages: $\mu = 24$, $\sigma^2 = 430$. How large must n be so $\bar{X}_n$ is within \$1 of μ with 99% probability?

$$\mathbb{P}[|\bar{X}_n - \mu| \geq 1] \underset{a}{\sim} \mathbb{P}[|Z| \geq \sqrt{n/430}] = .01$$

$$\sqrt{n/430} = 2.576 \quad \Rightarrow \quad n = 430 \times 2.576^2 \geq 2,862$$

Compare with Chebyshev: $n = 43,000$. The CLT is **15× more efficient** because it uses the shape of the distribution, not just the variance.

CLT in Action: R Simulation

```
# CLT: sample means of Exponential(1) converge to Normal
set.seed(42)
par(mfrow = c(2, 2))
for (n in c(5, 30, 100, 1000)) {
  xbar <- replicate(10000, mean(rexp(n, rate = 1)))
  hist(xbar, breaks = 40, prob = TRUE,
       main = paste("n =", n),
       xlab = expression(bar(X)[n]))
  curve(dnorm(x, 1, 1/sqrt(n)), add = TRUE,
        col = "red", lwd = 2)
}
```

The red curve is the normal approximation $N(\mu, \sigma^2/n)$. Even for **skewed** distributions (Exponential), the normal approximation is excellent by $n = 30$.

When Do Asymptotics Kick In?

Data shape	Rule of thumb	Example
Symmetric, light tails	$n \geq 30$	Heights, test scores
Moderate skew	$n \geq 100$	Wages, income
Heavy tails / outliers	$n \geq 500+$	Financial returns, conflict
Rare binary outcome	Events ≥ 10 /covariate	Coups, wars

There is no universal rule. The CLT convergence rate depends on the **shape** of the underlying distribution. When in doubt, **simulate**.

Multivariate CLT

The CLT generalizes to vectors. If $\mathbf{W}_1, \dots, \mathbf{W}_n$ are iid k -vectors with $\mathbb{E}[\mathbf{W}_i] = \boldsymbol{\mu}$ and $\text{Var}(\mathbf{W}_i) = \boldsymbol{\Sigma}$, then:

$$\sqrt{n}(\bar{\mathbf{W}}_n - \boldsymbol{\mu}) \xrightarrow{d} N(\mathbf{0}, \boldsymbol{\Sigma})$$

Applications:

- **OLS** (Lecture 11): $\frac{1}{\sqrt{n}} \sum_{i=1}^n X_i e_i \xrightarrow{d} N(\mathbf{0}, \sigma^2 Q_{XX})$
- **Probit MLE** (Lecture 9): $\frac{1}{\sqrt{n}} S_n(\beta_0) \xrightarrow{d} N(\mathbf{0}, \mathcal{I}(\beta_0))$
- **GMM** (coming later): $\frac{1}{\sqrt{n}} \sum_{i=1}^n g(W_i, \theta_0) \xrightarrow{d} N(\mathbf{0}, \Omega)$

In each case, a sum of mean-zero random vectors converges to a multivariate normal. The covariance matrix differs, but the **structure** is identical.

What We Developed Today

Tool	Statement	Application
WLLN	$\bar{X}_n \xrightarrow{P} \mu$	Consistency
CMT	$\text{plim } g(X_n) = g(\text{plim } X_n)$	Functions of estimators
O_p, o_p	Rate of convergence	Tracking remainders
CLT	$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2)$	Asymptotic normality

The CLT proof brought together:

- 1 Taylor expansion of the MGF (second-order approximation)
- 2 $O(\cdot)$ notation to track the remainder
- 3 The exponential limit $(1 + x/n)^n \rightarrow e^x$
- 4 MGF continuity theorem to conclude convergence in distribution

Looking Ahead

Lecture 11 introduces the **Delta Method** and applies all of today's tools to OLS:

- **Delta Method:** Taylor expansion + CLT $\Rightarrow$ distribution of $h(\hat{\beta})$
- **Consistency:** WLLN $\Rightarrow \frac{1}{n}\mathbf{X}'\mathbf{X} \xrightarrow{P} Q_{XX}$, CMT $\Rightarrow \hat{\beta} \xrightarrow{P} \beta$
- **Asymptotic normality:** Multivariate CLT $\Rightarrow \sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N(0, V_\beta)$
- **Wald tests:** Delta method $\Rightarrow$ test nonlinear hypotheses about β

These tools also justify what we did informally in Lecture 9 (Probit):

- Score CLT: $\frac{1}{\sqrt{n}}S_n(\beta_0) \xrightarrow{d} N(0, \mathcal{I}(\beta_0))$ (multivariate CLT)
- Hessian convergence: $\frac{1}{n}H_n(\beta_0) \xrightarrow{P} -\mathcal{I}(\beta_0)$ (WLLN)
- Combining via Slutsky: $\sqrt{n}(\hat{\beta} - \beta_0) \xrightarrow{d} N(0, \mathcal{I}^{-1})$